

UI Test Harness — What Changed

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What we set out to do

Until now, the only way to be confident a change to the bookkeeper-facing screens (Invoices and Submission Queue) hadn't broken something was to click through the app by hand. That's slow, easy to skip when deadlines are tight, and it's how the kind of subtle counting bug behind ticket TRB-011 slip through.

We added a lightweight, automated way to exercise those screens — the same way a real bookkeeper would click them — and check the results in seconds, every time anyone changes the code.

What's different now

- Every code change automatically re-runs a small suite of UI checks before it can be merged.
- Three concrete bookkeeper-experience scenarios are now guarded by these checks (details below).
- The new tests run in about five seconds. They were added alongside the 346 existing back-end tests so the whole suite still finishes in one command.
- No new costs or services. Everything runs on the developer's laptop and in the existing build pipeline.

The three scenarios we locked down

1. Bulk “Queue everything” respects the 8-invoice filing cap

If a bookkeeper has 10 ready invoices for one customer and clicks the "Send all eligible invoices to queue" button, the system queues only 8 (the per-filing limit the State portal can handle in one go) and tells the user which customer got skipped. We now have an automated check that fails the moment that behavior regresses.

2. Individual “Queue” button refuses to over-fill a customer

If a customer already has 8 invoices queued, the "Queue for Submission" button on a 9th invoice shows an immediate error (“Queue full (8 of 8)”) and does NOT send a partial write to the database. The new check confirms that no database write is attempted in that case.

3. The “N of 8 queued” counter changes color when the limit is reached

When a customer has 5 invoices queued, the counter shows “5 of 8 queued” in normal grey type. When it hits 8, the line switches to amber and bold so the bookkeeper sees they're at the maximum before clicking submit. Both states are now verified automatically.

Why this matters

- Tickets like TRB-011 — where the queue button was over-filling submissions and forcing manual cleanup — are exactly the class of bug these checks catch.
- The 8-invoice cap exists because the State of Arkansas's portal has a 15-minute time-out per filing and can't reliably process more in one batch. Breaking the cap means real, customer-visible

failures (the May 8 incident, submission aca53960, ran 48 minutes and was killed at row 27 of 29). The new checks make sure that limit can't silently slip.

- Future UI changes that touch these screens will fail this suite the moment they break a behavior — before the change reaches a customer.

Scope of this work

This is the test infrastructure plus three seed scenarios. It does NOT yet cover every screen in the product. As we touch other screens we'll add similar checks for them; the heavy lift of standing up the framework is now done.

Verification

- 346 existing back-end tests: still passing.
- 4 new UI tests: passing.
- Production build: passes, no extra weight added to what customers download.
- No production code was changed — only test scaffolding was added.

Generated from the ui-test-harness-plan deliverable. Technical detail (file list, configuration, exact test names) lives in the engineering README.